



LEADING UNIVERSITY

Department of Computer Science and Engineering

Course Code: CSE-4116

**Project Title: Build a Responsive Web Application with User Authentication
and Profile Management.**

Submitted To:

Somapika Das

Lecturer

Department of Computer Science and Engineering

Leading University, Sylhet

Submitted By:

Mst Jubaida Begum

ID:0182310012101106

Batch:61 Section: C

Department of Computer Science and Engineering

Date of Submission

24-05-2026

Recommendation Letter from the Project Supervisor

The project entitled "*Build a Responsive Web Application with User Authentication and Profile Management.*"

submitted by the student

Mst Jubaida Begum

0182310012101106

This is a record of research work carried out under my supervision, and I hereby approve that the report be submitted in partial fulfillment of the requirements for the award of the bachelor's degrees.

Somapika Das

Date: 24th May, 2026

Certificate of Acceptance of the Project

The project entitled "Build a Responsive Web Application with User Authentication and Profile Management."

submitted by the student

Mst Jubaida Begum

0182310012101106

on 24th May, 2026

is hereby accepted as the partial fulfillment of the requirements for the award of the bachelor's degrees.

Abstract

This project presents the design and development of a fully responsive web application that enables users to register, log in, and manage their personal profiles through a secure and user-friendly interface. The system is developed using modern web technologies including HTML, CSS, Bootstrap, and JavaScript for the front-end, and PHP with MySQL for the back-end. The application implements complete user authentication with email verification using a token-based mechanism, secure session management, and full CRUD (Create, Read, Update, Delete) functionality for user profile data, including profile image upload and management.

Additional features include a dummy payment simulation inspired by popular Bangladeshi payment platforms, a password reset mechanism using PHPMailer, a dark/light mode toggle, and a user activity log that records login history and account actions. The system also provides real-time popup notifications and validation feedback to enhance user experience and usability. Furthermore, the application is fully responsive and optimized for both desktop and mobile devices using Bootstrap and custom CSS techniques.

The project demonstrates the practical implementation of modern web development practices, secure authentication systems, responsive user interface design, and database-driven dynamic content management within a clean, maintainable, and scalable architecture. The application was developed and tested in a local XAMPP environment, while version control and project management were maintained using GitHub.

Index Terms- PHP, MySQL, Authentication, CRUD Operations, Bootstrap, Session Management, Responsive Web Application, PHPMailer, Email Verification, User Management System.

Acknowledgement

The successful completion of the *Build a Responsive Web Application with User Authentication and Profile Management* project represents a significant milestone in our academic journey. This achievement would not have been possible without the guidance and support of several individuals and institutions.

I express my sincere gratitude to my respected instructor, Somapika Das, for her continuous supervision, valuable advice, and thoughtful direction throughout the project. Her mentorship helped me refine my ideas and develop the system in a structured and effective manner.

I am also thankful to the Department of Computer Science and Engineering, Leading University, for providing the academic knowledge and technical foundation necessary to accomplish this work successfully.

Contents

Abstract.....	I
Acknowledgement.....	II
Table of Contents.....	III
Chapter 1: Introduction.....	1
Chapter 2: Problem Description.....	2
2.1 Problem Statement.....	2
2.2 Goals.....	2
2.3 Purposes.....	3
2.4 Methods.....	3
2.5 Related Work.....	3
Chapter 3: Requirement Specification.....	5
3.1 Functional Requirements.....	5
3.2 Nonfunctional Requirements.....	5
3.3 System Diagram	6
Chapter 4: Accomplishment.....	14
4.1 Preliminaries.....	14
4.2 How Work Was Done.....	15
Chapter 5: Results.....	20
Chapter 6: Conclusion.....	23
6.1 Restrictions.....	23
6.2 Limitations.....	23
6.3 Future Work.....	24
References.....	25
Appendices.....	26
Appendix A: Source Code.....	26
Appendix B: User Guide.....	27

Chapter 1: Introduction

The rapid growth of the internet has made web-based applications an essential part of modern daily life. From banking and e-commerce platforms to social media and government services, web applications serve as the primary medium through which users access information, manage data, and interact with digital systems. Among the most fundamental features that any modern web application must provide are user registration, secure authentication, and profile management — the core building blocks of a personalized and secure digital experience.

This project, titled “Build a Responsive Web Application with User Authentication and Profile Management,” was undertaken as the final project for the course Web Technologies (CSE 4116) under the Department of Computer Science and Engineering at Leading University. The primary objective of this project is to design, develop, and implement a complete, functional, and responsive web application that enables users to securely register using a validated email address, log in with authenticated credentials, and manage their personal profiles through a clean, modern, and intuitive interface.

The system is developed using a standard LAMP-inspired technology stack consisting of HTML5, CSS3, Bootstrap 5, and JavaScript for the client-side interface, while PHP 8 and MySQL are used for server-side processing and database management. The application is deployed and tested in a local XAMPP environment. The project demonstrates full CRUD (Create, Read, Update, Delete) functionality, session-based authentication, token-based email verification, image upload and management, dummy payment simulation, password reset functionality, dark/light mode toggling, and a complete user activity logging system.

In addition to functionality, significant emphasis has been placed on responsive user interface design, security, usability, and maintainability. The application is optimized for both desktop and mobile devices using Bootstrap’s responsive framework and custom CSS styling techniques.

This project not only fulfills the academic requirements of the course but also provides practical experience in full-stack web application development, secure authentication systems, database integration, and responsive user interface design. The completed system demonstrates how modern web technologies can be combined to create secure, scalable, and user-friendly web applications.

Chapter 2: Problem Description

2.1 Problem Statement

In many educational and small-scale organizational contexts, there is a need for a lightweight, self-hosted web application that allows individuals to create accounts, authenticate securely, and manage their personal information. Existing platforms are often too complex, dependent on third-party services, or require paid infrastructure. The challenge is to build a complete, self-contained web application that handles user registration, secure login, profile management with image upload, and additional utility features while minimizing dependency on complex third-party infrastructures or paid deployment services, while remaining fully responsive, secure, and academically sound.

2.2 Goals

The primary goals of this project are as follows:

- To design and implement a secure user registration and login system with front-end and back-end validation.
- To develop a responsive and accessible user interface compatible with both desktop and mobile devices.
- To implement full CRUD operations for user profile data, including profile image upload and deletion.
- To provide token-based email verification using PHPMailer and Gmail SMTP authentication.
- To include additional features such as payment simulation, password reset, dark mode, and activity logging.
- To ensure the codebase is clean, well-documented, and safely version-controlled on GitHub.

2.3 Purposes

The purpose of this project is to apply the theoretical knowledge gained throughout the Web Technologies course to a practical, real-world development scenario. It serves to demonstrate competency in full-stack web development using PHP and MySQL, understanding of authentication and session management, database design and CRUD operations, responsive UI design, and secure coding practices. The project also prepares the developer for industry-level web application development by simulating a complete development lifecycle from planning to deployment.

2.4 Methods

The application was developed following a structured approach. First, the database schema was designed and implemented in MySQL through phpMyAdmin running on XAMPP. The back-end logic was written in PHP using MySQLi for database interaction. The front end was built with Bootstrap 5 for responsiveness, custom CSS for visual design, and JavaScript for client-side validation and interactivity.

Email verification was implemented using PHPMailer with Gmail SMTP authentication and secure token-based verification links stored in the database. Additional features such as password reset, dummy payment simulation, dark/light mode toggling, and activity logging were integrated to enhance functionality and user experience.

All features were developed iteratively and tested locally before being version-controlled using Git and pushed to a GitHub repository.

2.5 Related Work

Modern authentication systems commonly utilize password hashing, session management, and email verification mechanisms to improve application security and user trust. Large-scale platforms such as Facebook, Google, GitHub, and LinkedIn use secure authentication systems that include email verification, password recovery, session handling, and profile management functionalities. These systems demonstrate the importance of secure user authentication and responsive user interfaces in modern web applications.

Web-based user authentication and profile management systems have been widely studied and implemented in both academic and commercial environments. Bootstrap-based responsive design has become a standard approach for ensuring cross-device compatibility in modern web applications [1]. PHP and MySQL remain among the most widely used technologies for dynamic web development in academic institutions, small businesses, and lightweight commercial applications [2].

Many educational management systems, online learning portals, and e-commerce applications also implement CRUD operations, profile image management, activity logging, and token-based verification systems similar to those developed in this project. Token-based email verification is a widely adopted authentication mechanism used to validate user identity and enhance account security in modern web applications.

The proposed system follows similar authentication and profile management concepts while focusing on simplicity, responsiveness, maintainability, and academic implementation using core web technologies.

Chapter 3: Requirement Specification

3.1 Functional Requirements

The following functional requirements define what the system must do:

- **User Registration:** The system shall allow new users to register using their full name, a real email address (validated using MX DNS record checking), and a password. Only one account shall be permitted per email address.
- **Email Verification:** Upon registration, a secure token-based verification link shall be generated and sent to the user's email address using PHPMailer and Gmail SMTP authentication. The user must click the link to activate their account before logging in.
- **User Login:** Registered and verified users shall be able to log in using their email and password. The system shall validate credentials against the database using hashed password comparison.
- **Session Management:** The system shall maintain user sessions after login and restrict access to protected pages for unauthenticated users.
- **Profile Management (CRUD):** Logged-in users shall be able to view, update, and delete their profile data, including full name, email address, and profile image. Profile images shall be uploaded from the user's local device and stored on the server.
- **Password Reset:** Users shall be able to request a password reset via a token-based link and set a new password.
- **Dummy Payment:** Users shall be able to simulate a payment transaction using card information and view payment-related activity.
- **Dark/Light Mode:** Users shall be able to toggle between dark and light display modes for improved user experience.
- **Activity Log:** The system shall record and display all significant user actions, including login, logout, profile update, and payment attempts.
- **Popup Messages:** All operations, including registration, login, update, delete, and payment, shall display real-time popup alert messages indicating success or failure.

3.2 Non-functional Requirements

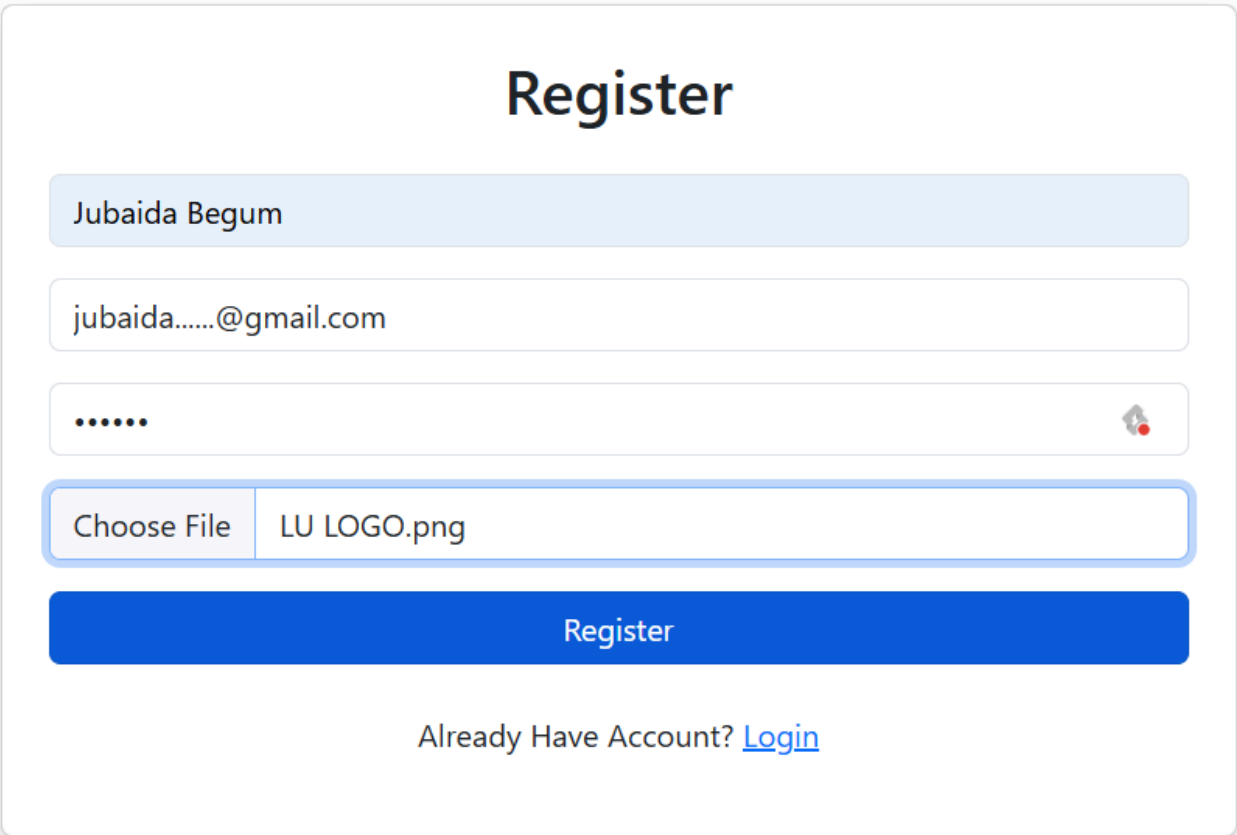
- **Responsiveness:** The application shall be fully responsive and usable on screen sizes ranging from mobile (320px) to large desktop (1920px) using Bootstrap 5.
- **Security:** All passwords shall be hashed using PHP's password_hash() function. Basic input validation and password hashing mechanisms shall be implemented to improve application security. Session data shall be validated on every protected page.
- **Usability:** The interface shall be clean, intuitive, and require no technical knowledge to operate.

- **Performance:** Pages shall load within an acceptable time on a local XAMPP server without performance optimization requirements.
- **Maintainability:** The code shall be well-structured, commented, and organized into logical files and folders for ease of maintenance.
- **Portability:** The application shall run on any machine with XAMPP installed without requiring changes to the core codebase.
- **Git Safety:** No sensitive credentials, API keys, or passwords shall be committed to the GitHub repository.

3.3 System Diagrams

This section presents screenshots of the developed system that visually demonstrate the structure, flow, and interface of the application.

Figure 3.1 - Registration Page



The screenshot displays a registration form titled "Register". The form includes the following fields and elements:

- A text input field containing the name "Jubaida Begum".
- A text input field containing the email address "jubaida.....@gmail.com".
- A password input field represented by six dots, with a small eye icon on the right for toggling visibility.
- A file upload section with a "Choose File" button and the filename "LU LOGO.png".
- A prominent blue "Register" button.
- A link at the bottom that reads "Already Have Account? [Login](#)".

Figure 3.1: Registration page showing user registration form with validation.

Figure 3.2 - Login Page

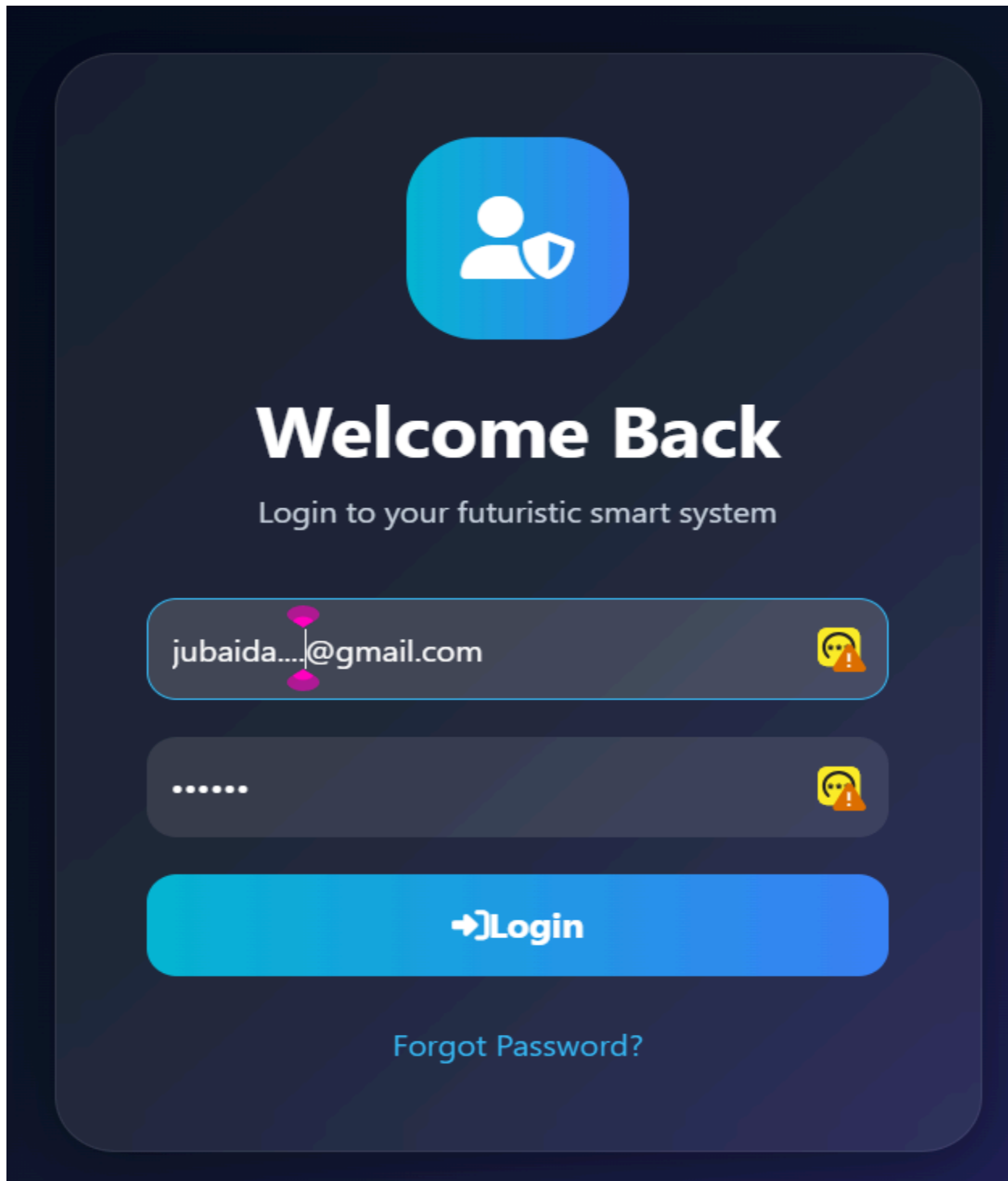


Figure 3.2: Secure login page for authenticated user access

Figure 3.3 - User Dashboard



Figure 3.3: Main user dashboard with profile overview and quick navigation

Figure 3.4 - Profile Management Page

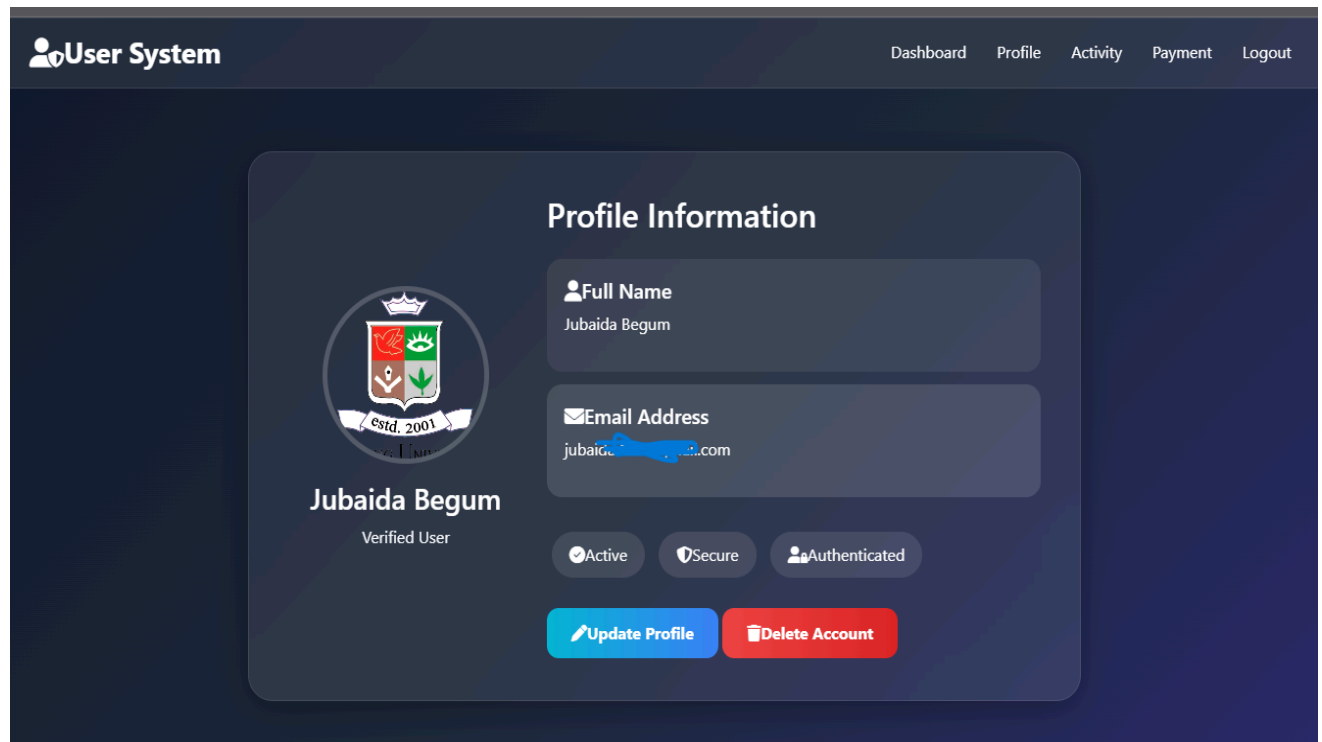


Figure 3.4: Profile management page for updating user information and image.

Figure 3.5 -Activity log

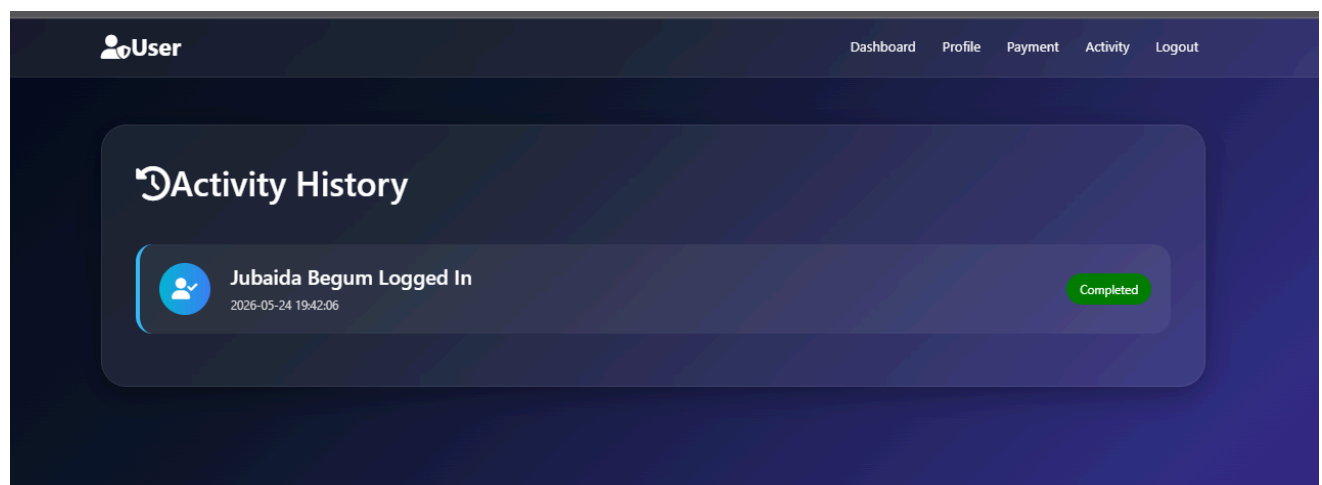
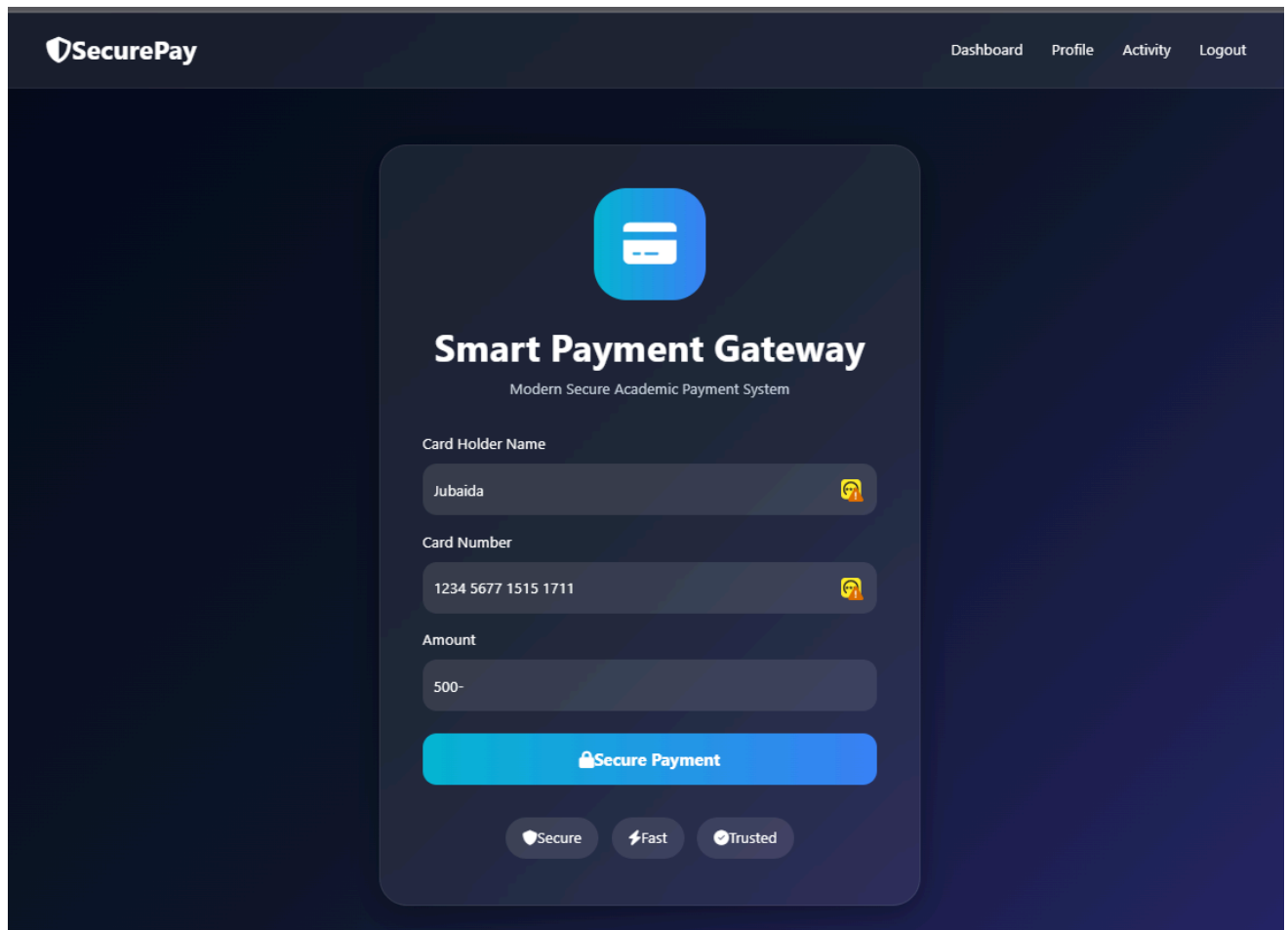


Figure 3.5:Activity log page displaying user actions and system history.

Figure 3.6 -Dummy Payment Simulation



The image shows a web interface for 'SecurePay' with a dark blue background. At the top, there is a navigation bar with the 'SecurePay' logo on the left and links for 'Dashboard', 'Profile', 'Activity', and 'Logout' on the right. The main content area features a central white card with rounded corners. At the top of the card is a blue square icon with a white credit card symbol. Below the icon, the text 'Smart Payment Gateway' is displayed in a bold, sans-serif font, followed by the subtitle 'Modern Secure Academic Payment System' in a smaller font. The card contains three input fields: 'Card Holder Name' with the value 'Jubaida', 'Card Number' with the value '1234 5677 1515 1711', and 'Amount' with the value '500-'. Each input field has a small yellow icon with a person silhouette to its right. Below the input fields is a large, rounded rectangular button with a blue-to-white gradient, labeled 'Secure Payment' with a lock icon. At the bottom of the card, there are three small, rounded buttons: 'Secure' with a shield icon, 'Fast' with a lightning bolt icon, and 'Trusted' with a checkmark icon.

SecurePay

Dashboard Profile Activity Logout

Smart Payment Gateway
Modern Secure Academic Payment System

Card Holder Name
Jubaida

Card Number
1234 5677 1515 1711

Amount
500-

Secure Payment

Secure Fast Trusted

Figure 3.6:Dummy payment simulation page with transaction processing interface.

Figure 3.7 -Delete Page

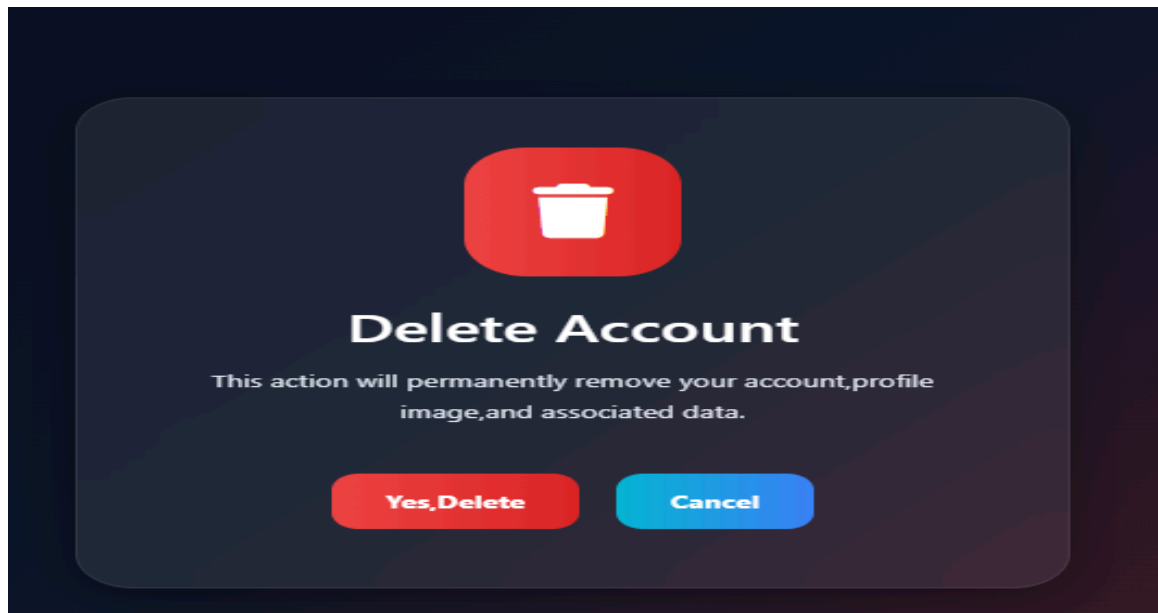


Figure 3.7: Account deletion page with secure confirmation system.

Figure 3.8 -Logout Page

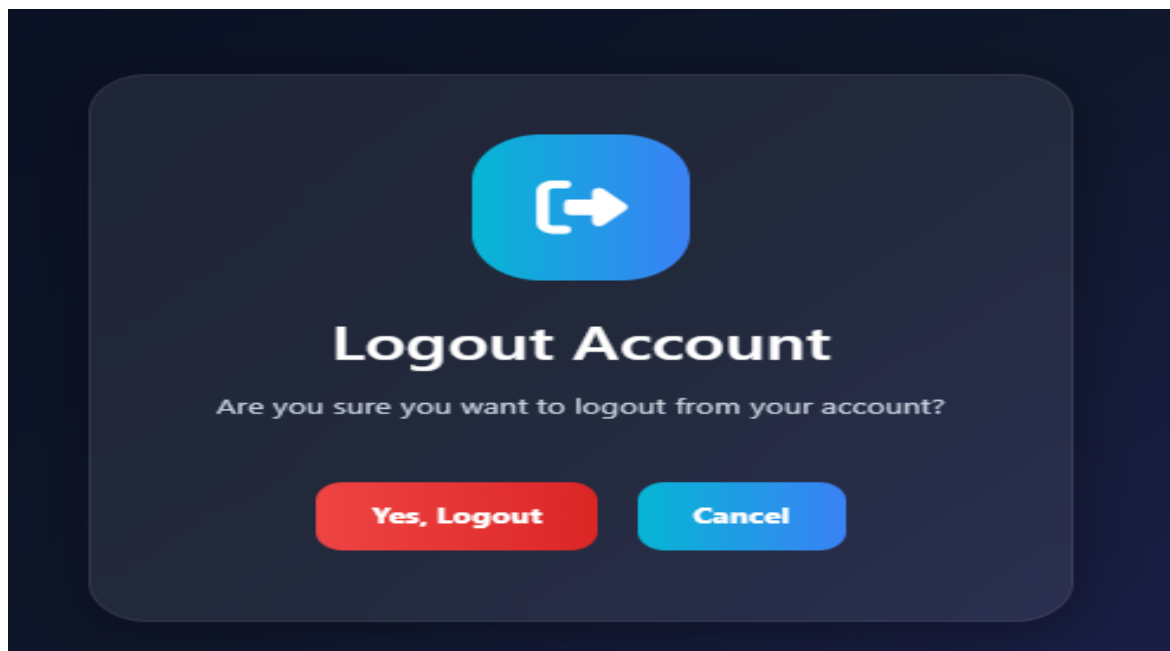


Figure 3.8: Logout confirmation system for secure session termination.

Figure 3.9 -Theme Toggling Page



Figure 3.9: Dark and light mode interface demonstrating responsive UI design.

Figure 3.10 -Password Reset Page

The image displays two mobile application screens for password management, set against a dark blue background.

Left Screen: Forgot Password

- Icon: A blue padlock.
- Title: **Forgot Password**
- Subtitle: Generate your secure password reset link
- Input Field: "Enter Your Email..." with a yellow envelope icon on the right.
- Button: "Send Reset Link" with a blue-to-white gradient and a white arrow icon.
- Side Button: "Open Reset Password Page" (blue button on the left edge).

Right Screen: Reset Password

- Icon: A blue rounded square containing a white key.
- Title: **Reset Password**
- Subtitle: Create your new secure password
- Section Header: **New Password**
- Input Field: "Enter New Password" with a red and white shield icon on the right.
- Button: "Reset Password" with a blue-to-white gradient and a white shield icon.

Figure 3.10: Password reset page using token-based authentication.

Chapter 4: Accomplishment

Accomplishment

This project represents a significant milestone in digital educational tools, successfully transitioning from a conceptual framework to a fully synchronized, data-driven mobile/web platform for a user management system.

4.1 Preliminaries

The development of this project began with careful planning, environment configuration, and technology selection to ensure that the application would satisfy all project requirements efficiently and professionally.

Development Environment Setup

The project was developed and tested in a local XAMPP environment on a Windows operating system. XAMPP provided the Apache web server, PHP 8 runtime environment, and MySQL database server required for full-stack web application development.

Database Configuration

The database was designed and managed using phpMyAdmin. Multiple relational tables were created to manage system functionality, including:

- users table for storing user account information
- activity_logs table for storing user activities and login history
- payments table for storing dummy payment transaction records

The database structure was normalized to ensure organized and maintainable data management.

Front-End Development Tools

Visual Studio Code was used as the primary development environment. Bootstrap 5 was integrated through CDN to provide a fully responsive layout system compatible with desktop and mobile

devices. Custom CSS and JavaScript were additionally implemented to enhance visual appearance, animations, validation, and user interaction.

Version Control and Project Management

Git was used for version control throughout the development process. The project source code was managed using a GitHub repository, allowing safe backup, organized version tracking, and structured project maintenance. Sensitive information such as passwords and configuration files were excluded from public commits using a .gitignore file.

Project Structure

The project files were organized modularly into separate folders for authentication, database connection, uploads, styling, and reusable includes. This structure improved maintainability, readability, and future scalability of the application.

The application was stored inside the XAMPP htdocs directory and executed locally using:

`http://localhost/user-management-system/`

4.2 How the Work Was Done

The project was implemented following a structured and feature-oriented development approach. Each major module was developed separately and integrated progressively into the complete system.

Phase 1: Database Design and Authentication System

The initial phase focused on designing the database architecture and implementing secure user authentication functionality.

The registration system was developed using HTML forms, Bootstrap styling, JavaScript validation, and PHP back-end processing. User passwords were securely encrypted using PHP's `password_hash()` function before being stored in the database.

To improve security and prevent invalid registrations, email validation was implemented both on the client side and server side. The system checks valid email formatting using JavaScript and verifies real email domains using PHP's `checkdnsrr()` function with MX record checking.

A token-based email verification mechanism was implemented using PHPMailer and Gmail SMTP authentication. After successful registration, the system sends a verification email containing a unique activation link generated through secure random token creation. Users must verify their email before being allowed to log in.

The login system authenticates users using `password_verify()` and establishes secure PHP sessions upon successful authentication.

Phase 2: Session Management and Protected Routing

Session-based authentication was implemented to maintain secure user access throughout the application.

When a user successfully logs in, important user information is stored inside PHP session variables. Every protected page verifies session validity before allowing access. Unauthorized users attempting to access restricted pages are automatically redirected to the login page.

Logout functionality destroys all session data securely and records the logout activity in the activity log system.

Phase 3: Profile Management and CRUD Operations

The profile management module was developed to support complete CRUD (Create, Read, Update, Delete) functionality.

Users are able to:

- View their profile information
- Update personal information
- Upload profile images
- Delete their account permanently

Profile images are uploaded from the user's local device using HTML file input fields. JavaScript-based image preview functionality was implemented to improve user experience before upload.

Uploaded images are stored inside the server upload directory, while image filenames are saved inside the database. During account deletion, associated profile images are removed from the server to avoid unused file accumulation.

Dynamic PHP queries were used to update and retrieve user data efficiently from MySQL.

Phase 4: Responsive User Interface Development

Significant attention was given to modern UI/UX design and responsiveness.

Bootstrap 5 grid systems and utility classes were extensively used to ensure compatibility across mobile phones, tablets, laptops, and desktop devices.

Custom CSS styling was implemented to create:

- Glassmorphism effects
- Gradient backgrounds
- Animated hover effects
- Modern dashboard cards
- Smooth transitions and shadows
- Dark and light mode appearance

Responsive navigation bars, cards, profile sections, and forms were carefully optimized for smaller screen sizes using media queries.

Phase 5: Additional Feature Integration

Several additional features were implemented to enhance the functionality and uniqueness of the project.

Dummy Payment Simulation

A payment simulation module was created where users can simulate transactions using payment methods.

The system generates random transaction IDs and displays success or failure popup alerts to imitate real payment gateway behavior.

Password Reset System

A secure password reset mechanism was implemented using token-based verification links. Users can request password recovery and securely create a new password.

Dark and Light Mode Toggle

A dark/light mode switching system was developed using JavaScript and CSS class toggling. The selected theme dynamically changes the interface appearance to improve accessibility and user preference.

Activity Logging System

A complete user activity logging system was implemented. Important actions such as:

- Registration
- Login
- Logout
- Profile Update
- Payment Attempts
- Password Reset

are automatically stored in the database with timestamps for tracking and monitoring purposes.

4.3 Key Technical Milestones

The following major accomplishments were achieved during the project implementation:

- Successful implementation of secure registration and login functionality using PHP sessions and hashed passwords.
- Integration of PHPMailer with Gmail SMTP for token-based email verification.
- Development of complete CRUD functionality for user profile management.
- Implementation of responsive modern UI using Bootstrap 5 and custom CSS effects.
- Creation of a functional dummy payment simulation system.
- Integration of secure password reset functionality.
- Successful implementation of dark/light mode switching using JavaScript.
- Development of a user activity logging system for monitoring important user actions.
- Proper GitHub version control and organized project structure for maintainability.
- Successful testing across desktop and mobile screen sizes to ensure full responsiveness.

Chapter 5: Results

The implementation phase successfully produced a complete, responsive, and secure web application that integrates user authentication, profile management, session handling, and additional modern web features into a unified system. The results are categorized into functional outcomes and system performance.

5.1 Responsive User Interface and Navigation Performance

The application successfully delivers a modern, responsive, and visually interactive user interface across desktop and mobile devices. By utilizing Bootstrap 5 grid systems, custom CSS styling, glassmorphism effects, and responsive media queries, the system achieves smooth navigation and adaptive layouts across different screen sizes.

Outcome:

Users experience smooth interaction throughout the application, including navigation between registration, login, dashboard, profile management, payment, and activity log pages. Hover animations, gradient backgrounds, popup alerts, and responsive cards provide a modern user experience while maintaining usability and accessibility.

Technical Result:

The responsive layout automatically adjusts content, forms, navigation bars, profile sections, and dashboard cards for mobile, tablet, and desktop displays without breaking the interface structure.

5.2 Authentication and Session Management Results

A major objective of the project was the implementation of a secure authentication and session management system.

Result:

The registration and login modules successfully authenticate users using encrypted password hashing and session-based authentication. Email verification using PHPMailer and Gmail SMTP integration correctly activates user accounts before login access is granted.

Technical Proof:

Upon successful login, PHP session variables are created and maintained across all protected pages. Unauthorized users attempting to access restricted routes are automatically redirected to the login page, ensuring secure system access.

The activity logging mechanism records important actions such as:

- Registration
- Login
- Logout
- Password reset
- Profile updates
- Payment attempts

with timestamps stored inside the database.

5.3 Profile Management and CRUD Functionality

The profile management module successfully demonstrates complete CRUD (Create, Read, Update, Delete) operations using PHP and MySQL.

Result:

Users can successfully:

- View profile information
- Update personal details
- Upload profile images
- Delete their accounts permanently

Profile image uploads function correctly through local device selection and server-side storage.

Validation:

The system dynamically retrieves updated profile information from the database and displays changes immediately on the dashboard and profile pages. Uploaded images are previewed before submission and stored securely inside the server upload directory.

When an account is deleted, the corresponding database record and uploaded image file are removed successfully from the system.

5.4 Additional Feature Integration Results

Several additional features were successfully integrated to enhance system functionality and user experience.

Dummy Payment Simulation

The payment module successfully simulates online payment transactions using methods.

The system generates random transaction IDs and dynamically displays transaction success or failure messages through popup alerts.

Password Reset Functionality

The password reset system correctly generates secure token-based reset links and allows users to create new passwords securely. Invalid or expired reset tokens are rejected automatically.

Dark and Light Mode Toggle

The dark/light mode functionality successfully changes the application appearance dynamically using JavaScript and CSS class toggling, improving accessibility and user customization.

Error Handling and Validation

The application demonstrates stable behavior during invalid operations and edge cases. Incorrect passwords, duplicate emails, invalid verification tokens, and unauthorized page access are handled gracefully using real-time popup messages without crashing the system.

Chapter 6: Conclusion

This project successfully demonstrates the design and implementation of a complete, responsive web application with user authentication and profile management using PHP, MySQL, Bootstrap, JavaScript, and PHPMailer. All project requirements were successfully fulfilled, including user registration with validated email addresses, secure login with session management, full CRUD operations for profile management with image handling, email verification, password reset functionality, dummy payment simulation, dark/light mode toggle, and user activity logging.

The project provided valuable practical experience in full-stack web development, database design, responsive user interface development, authentication systems, session handling, and secure coding practices. It also demonstrated how modern web technologies can be integrated to build a secure, maintainable, and user-friendly web application suitable for academic and small-scale organizational use.

The application successfully combines functionality, responsiveness, and modern UI design principles within a structured and maintainable codebase developed in a local XAMPP environment with GitHub-based version control.

6.1 Restrictions

The application is currently configured to run within a local XAMPP development environment and has not yet been deployed to a public production server. As a result, access to the system is limited to machines where the required local server environment is properly configured.

Additionally, Gmail SMTP authentication requires application-specific passwords and proper security configuration, which may vary depending on deployment environments and Google account policies.

6.2 Limitations

- The application currently operates in a local development environment and has not undergone large-scale production testing.
- The dummy payment module simulates transactions only and does not connect to real payment gateway APIs.
- The system does not currently implement advanced protection mechanisms such as CAPTCHA or rate limiting against brute-force login attempts.
- Uploaded image validation is primarily extension-based; stronger MIME-type validation and image sanitization could further improve security.
- Email delivery depends on external SMTP configuration and internet connectivity for successful verification email transmission.

6.3 Future Work

The project can be further improved and extended through the following enhancements:

- Deployment to a live production server using shared hosting or cloud platforms such as AWS, DigitalOcean, or cPanel hosting.
- Integration of real payment gateway APIs such as bKash, Nagad, Stripe, or PayPal.
- Implementation of two-factor authentication (2FA) for improved account security.
- Addition of advanced security features including CAPTCHA, login rate limiting, and CSRF protection.
- Integration of OAuth-based social login systems such as Google or GitHub authentication.
- Development of a progressive web application (PWA) or dedicated mobile application version for enhanced accessibility and performance.

References

- [1] M. Otto and J. Thornton, “Bootstrap Documentation,” Bootstrap, 2026. [Online]. Available: <https://getbootstrap.com/>
- [2] PHP Documentation Team, “PHP: Hypertext Preprocessor,” PHP Official Documentation, 2026. [Online]. Available: <https://www.php.net/>
- [3] Oracle Corporation, “MySQL Documentation,” MySQL Official Documentation, 2026. [Online]. Available: <https://dev.mysql.com/doc/>
- [4] PHPMailer Team, “PHPMailer GitHub Repository,” GitHub, 2026. [Online]. Available: <https://github.com/PHPMailer/PHPMailer>
- [5] Mozilla Developer Network (MDN), “JavaScript Guide,” MDN Web Docs, 2026. [Online]. Available: <https://developer.mozilla.org/>
- [6] Apache Friends, “XAMPP Installation and Documentation,” Apache Friends, 2026. [Online]. Available: <https://www.apachefriends.org/>
- [7] GitHub Inc., “GitHub Documentation,” GitHub Docs, 2026. [Online]. Available: <https://docs.github.com/>

Appendices

Appendix A

Source Code

Appendices are included to provide supplementary technical materials that support the implementation and evaluation of the project. In this project, Appendix A presents an overview of the application structure, source code organization, and implementation methodology used in the development of the responsive user management system.

The project follows a modular PHP-based architecture to ensure maintainability, readability, and separation of concerns. The major modules of the system include:

- **Authentication Module** – Handles user registration, login, logout, session validation, email verification, and password reset functionality.
- **Dashboard Module** – Displays user information, quick-access navigation, and personalized interface components.
- **Profile Management Module** – Supports full CRUD operations for user profile data, including profile image upload, update, and account deletion.
- **Payment Simulation Module** – Simulates payment transactions using methods such as bKash, Nagad, VISA, and MasterCard with transaction history support.
- **Activity Log Module** – Records user actions such as login, logout, profile updates, and payment attempts with timestamps and IP addresses.
- **Theme Management Module** – Implements dark and light mode functionality with preference storage in the database.
- **Database Module** – Manages database connections and secure query execution using MySQLi prepared statements.
- **Validation Module** – Performs both client-side and server-side validation for secure data handling.

Key Implementation Highlights

- Passwords are securely hashed using PHP `password_hash()` and verified using `password_verify()`.
- Email verification and password reset are implemented using secure random token generation.
- JavaScript is used for real-time form validation, password matching, image preview, and interactive UI behavior.
- Bootstrap 5 responsive grid system ensures compatibility across desktop and mobile devices.
- User sessions are protected through centralized authentication checks on restricted pages.
- Profile images are uploaded to the local server and managed dynamically.
- All major user actions are stored in the activity log for tracking purposes.
- Git and GitHub were used for version control and project management.

The complete project source code is available in the GitHub repository for academic evaluation and demonstration purposes.

Appendix B

User's Guide

This section provides instructions for installing, configuring, and operating the developed web application.

B.1 Installation and Setup

Software Requirements

Before running the project, the following software must be installed:

- XAMPP (Apache + PHP + MySQL)
- Web Browser (Google Chrome / Microsoft Edge / Firefox)
- Visual Studio Code (optional for editing)
- Git (optional for version control)

Installation Steps

1. Install XAMPP on the system.
2. Start:
 - Apache Server
 - MySQL Server
3. Copy the project folder into:

C:\xampp\htdocs\

4. Open phpMyAdmin from:

<http://localhost/phpmyadmin>

5. Create a database for the project.
6. Import the provided SQL database file into phpMyAdmin.
7. Configure database credentials inside:

includes/db.php

8. Run the project in the browser:

<http://localhost/user-management-system>

B.2 User Registration

1. Open the registration page.
2. Enter:
 - Full Name
 - Email Address
 - Password
3. Submit the form.
4. The system validates:
 - Email format
 - Password matching

- Duplicate email existence
- 5. A verification link is generated.
- 6. Open the verification link to activate the account.

B.3 User Login

1. Open the login page.
2. Enter registered email and password.
3. Click “Login”.
4. If credentials are valid and the account is verified:
 - The user is redirected to the dashboard.
 - A session is created securely.
5. Invalid credentials display popup error messages.

B.4 Dashboard Access

After successful login, users can:

- View profile information
- Access payment simulation
- Open activity logs
- Update profile data
- Toggle dark/light mode
- Reset password
- Logout securely

The dashboard is responsive and accessible on both desktop and mobile devices.

B.5 Profile Management

Users can manage personal information through the profile section.

Available Operations

- View profile data
- Update:
 - Name
 - Email Address
 - Profile Image
- Delete account permanently

Image Upload

Users can:

1. Select an image from the local device.
2. Preview the image before upload.
3. Save the updated profile.

Uploaded images are stored on the server and displayed dynamically.

B.6 Password Reset

1. Open the “Forgot Password” page.
2. Enter registered email address.
3. A secure reset link is generated.
4. Open the link.
5. Enter a new password.
6. Submit to update the password securely.

B.7 Dummy Payment Simulation

The application includes a payment simulation feature for demonstration purposes.

Payment Flow

1. Enter your name.
2. Enter card number.
3. Enter amount

4. Submit payment.
5. The system:
 - Generates a transaction ID
 - Simulates success or failure
 - Stores payment history in the database

B.8 Dark/Light Mode

Users can switch between dark and light themes using the theme toggle button.

Features

- Real-time theme switching
- Smooth transition effects
- Preference stored in database
- Automatically applied after login

B.9 Activity Log

The system records important user activities including:

- Login
- Logout
- Profile updates
- Password reset
- Payment attempts
- Account deletion

Each record contains:

- Action type
- Timestamp
- User IP address

B.10 Logout

Users can securely log out using the logout button available in the navigation bar.

Upon logout:

- Session data is destroyed
- User is redirected to the login page
- Logout activity is recorded in the activity log table